Finite approximations of the localisation

a complement to *The localisation of the poset model*

21 October 2026

Abstract

For a type A of the poset model, $\mathsf{D}_N A$ is defined as $\mathsf{D}A$ (`localisation.tex`) with the families indexed only by the maps $[k] \to X$ with $k \leq N$. We show that $\mathsf{D}_N A = \mathsf{D}(K_N A)$, where $K_N A$ is the N-coskeleton of A: a type over the posets whose restriction to the linear posets is the simplicial coskeleton of i^*A. So $\mathsf{D}_N A$ is fibrant when A is, by Theorem 4.1 of `localisation.tex` applied to $K_N A$. Spheres of dimension at least N have unique fillers in $\mathsf{D}_N A$, and $\mathsf{D}A$ is the strict limit of the tower $(\mathsf{D}_N A)_N$; it is also the internal limit. The truncation level is shifted by one: $\mathsf{D}_0 A$ is the propositional truncation of A and $\mathsf{D}_1 A$ its set truncation, and in general D_{n+1} is the candidate for n-truncation. The maps of the tower are not fibrations.

1 The coskeleton

We use the notation of `localisation.tex`: $\mathcal{C}$ is the category of finite non-empty posets, $\Delta \subseteq \mathcal{C}$ the linear posets $[n] = \{0 < \cdots < n\}$, $\Delta_+ \subseteq \Delta$ the strictly monotone maps, and i^* the restriction to Δ.

Definition 1.1. Let $N \geq -1$ and let A be a type over Γ. For $\rho \in \Gamma(X)$, $K_N A(X, \rho)$ is the set of families $c(f) \in A([k], \rho f)$, for all $k \leq N$ and $f : [k] \to X$, such that $c(f)\, g = c(fg)$ for every $g : [j] \to [k]$ with $j \leq N$. Restriction along $X' \to X$ is precomposition. The map $\eta : A \to K_N A$ is $\eta(a)(f) = af$, and $K_M A \to K_N A$, for $N \leq M$, forgets the values above N.

The family and the condition are coded, so $K_N A$ is a type. On a linear poset, $K_N A([m], \tau)$ is the set of maps from the N-skeleton of $\Delta[m]$ to i^*A over τ: such a map is determined by its values on the simplices of dimension at most N, and these are natural along all maps. So $i^* K_N A$ is the N-coskeleton of i^*A. For $N = -1$ the families are empty and $K_{-1} A = 1$.

Lemma 1.2.

1. *For $m \leq N$, $\eta : A([m], \tau) \to K_N A([m], \tau)$ is a bijection, with inverse $c \mapsto c(\mathrm{id})$.*
2. *For $m > N$, restriction to the faces is a bijection from $K_N A([m], \tau)$ to the compatible families of elements of $K_N A([m-1], \tau d_i)$, $0 \leq i \leq m$.*
3. *A fibration structure on A gives a horn filling operation on $i^* K_N A$.*
4. *On a linear poset, A is the limit of the tower $\cdots \to K_{N+1} A \to K_N A \to \cdots \to K_{-1} A$.*

Proof. (1) For $m \leq N$ the identity of $[m]$ is one of the indices, and $c(g) = c(\mathrm{id})\, g$.

(2) A map $[k] \to [m]$ with $k \leq N < m$ misses a point of $[m]$, so it factors through a face d_i. If it factors through d_i and d_j, it factors through their intersection, on which compatible faces agree.

(3) Let h be a horn of i^*K_NA on $\Lambda^k[m+1]$. If $m \leq N$, its faces read at the identity are elements of $A([m])$, and two of them agree on their common face, by the naturality of h and (1). The construction of Lemma 2.1 uses only these faces and their agreement: it reads them on the poset horn and composes in A. Its filler a gives the filler $\eta(a)$. If $m > N$, a map $[j] \to [m+1]$ with $j \leq N$ misses at least two vertices, hence a vertex different from k, so it lies in the horn, and h read at these maps is the filler.

(4) By (1). □

The count in (3) is elementary: a monotone sequence on $[0, j]$ with values at most M and $j < M$ misses a value at most M; applied to the composite with the degeneracy at k, this gives a vertex $y \neq k$ missed by the map.

2 The approximations D_N

Definition 2.1. For a type A over Γ and $N \geq -1$, $\mathsf{D}_N A = \mathsf{D}(K_N A)$.

Proposition 2.2. *$\mathsf{D}_N A(X, \rho)$ is in bijection with the set of families $w(f) \in A([k], \rho f)$, for $k \leq N$ and $f : [k] \to X$, with $w(f)\, g = w(fg)$ for every strictly monotone $g : [m] \to [k]$. The bijection commutes with restriction.*

Proof. Given $u \in \mathsf{D}_N A(X, \rho)$, put $w(f) = u(f)(\mathrm{id})$; it is natural along strict maps, by the naturality of u and of $u(f)$. Conversely, given w, put $u(f)(g) = w(fm)\,\sigma$, where $g = m\sigma$ is the image factorisation of $g : [j] \to [k]$, m strict and σ surjective. Two factorisations are compared by a strict map μ (Eilenberg–Zilber): for g and gt, $\sigma t = \mu\sigma'$ and $m' = m\mu$, which gives the naturality of $u(f)$ along every t; for g and sg with s strict, $\sigma = \mu\sigma'$ and $m\mu = sm'$, which gives the naturality of u along s. Below N the value $w(fm)\sigma$ is $w(f)\, m\sigma = w(f)g$, so the two maps are inverse. □

So $\mathsf{D}_N A$ is $\mathsf{D}A$ with the indices restricted to $[0], \ldots, [N]$, and $\mathsf{D}A = \mathsf{D}_\infty A$.

Corollary 2.3. *If A is fibrant, so is $\mathsf{D}_N A$, naturally in Γ.*

Proof. By Lemma 1.2(3) and Theorem 4.1 applied to $K_N A$. Naturality is obtained on the universal family, as in Theorem 4.1: K_N preserves cartesian maps, as D does, so $\mathsf{D}_N(c^*\mathsf{El}) \to \mathsf{D}_N\mathsf{El}$ is cartesian over c, and these maps compose. □

Example 2.4.

- $\mathsf{D}_{-1}A = 1$.
- $\mathsf{D}_0 A(X, \rho) = \prod_{x \in X} A([0], \rho x)$, since the only strictly monotone map $[0] \to [0]$ is the identity. Any two elements u, v at X are the ends of a line, namely the element of $\mathsf{D}_0 A$ at $X \times [1]$ equal to u on $X \times \{0\}$ and to v on $X \times \{1\}$. So $\mathsf{D}_0 A$ is a proposition, with an element over ρ exactly when every $A([0], \rho x)$ has one: the propositional truncation of A, with no higher inductive type.
- $i^*K_1 A$ has the points and the edges of i^*A, and every compatible boundary of dimension ≥ 2 has a unique filler. Its connected components are those of i^*A, and its loops are all homotopic to the constant ones: $\mathsf{D}_1 A$ is the candidate set truncation.

Remark 2.5. The fibre of $\mathsf{D}_N A$ at $X = [0]$ is not the truncation of the fibre of A. It is the set of maps from the N-skeleton of the semi-simplicial set with one simplex in each dimension, which has the homotopy type of a sphere when N is odd. This does not matter, because the homotopy type of a D-modal type is read on its restriction to Δ.

3 Truncation level and limit

Proposition 3.1. *For $m > N$, restriction to the faces is a bijection from $\mathsf{D}_N A([m], \rho)$ to the compatible families of elements of $\mathsf{D}_N A([m-1], \rho d_i)$. So $i^*\mathsf{D}_N A$ has unique fillers for all spheres of dimension at least N.*

Proof. By Proposition 2.2, as in Lemma 1.2(2): a map $[k] \to [m]$ with $k \leq N < m$ misses a point, so it factors through a face. □

In a Kan complex with this property, the homotopy groups π_k vanish for $k \geq N$. A class in π_k is a k-simplex with boundary at the base point, and the sphere of dimension $k \geq N$ formed by this simplex and degenerate ones has a filler, which is a homotopy to the base point. Classically, $i^*A \to \mathrm{cosk}_N i^* A$ is moreover a bijection on π_k for $k < N$, since these use simplices of dimension at most $k+1 \leq N$ only. So $\mathsf{D}_{n+1}A$ is the candidate n-truncation of $\mathsf{D}A$.

Proposition 3.2. *$\mathsf{D}A$ is the limit of the tower $\cdots \to \mathsf{D}_{N+1}A \to \mathsf{D}_N A \to \cdots \to \mathsf{D}_{-1}A = 1$.*

Proof. By Proposition 2.2, a compatible sequence of elements of the $\mathsf{D}_N A(X, \rho)$ is a family indexed by all the maps $[k] \to X$, natural along the strictly monotone maps, that is, an element of $\mathsf{D}A(X, \rho)$. □

Remark 3.3. The maps of the tower are not fibrations. A line of $\mathsf{D}_0 A$ at $[0]$ is an element at $[1]$, that is, two points a, b of $A([0])$. A lift to $\mathsf{D}_1 A$ starting from a point of $\mathsf{D}_1 A$ over a gives an element of $\mathsf{D}_1 A([1])$, and in particular an edge of i^*A from a to b. If a and b lie in different components, there is none. The limit is therefore a strict limit, not the limit of a tower of fibrations. The same holds for the coskeletal tower of a Kan complex, which converges strictly although its maps are not Kan fibrations. Internally nothing is lost: every map is equivalent to the projection of the family of its fibres, so the internal tower $\mathsf{D}_{N+1}A \to \mathsf{D}_N A$ is a tower of fibrations up to equivalence. What does not hold automatically is that the strict limit computes the internal limit.

4 Convergence

Write $C = i^*A$ and $C_N = i^* K_N A$, the N-coskeleton of C, so that $\mathsf{D}_N A = \mathsf{D}(K_N A)$. Each C_N has a horn filling operation (Lemma 1.2(3)). In dimensions $\leq N$ the simplices of C_N are those of C, and in higher dimensions they are determined by their boundaries. For a poset P we identify P with its nerve, so that $\Delta[m] \times \Delta[1]^k$ is the poset $[m] \times [1]^k$. A map $P \to C_N$ over ρ is an element of $K_N A(P, \rho)$, since both are the maps from the N-skeleton of P to C.

Lemma 4.1. *Let T be a simplicial set with a horn filling operation, P a finite poset, $S \subseteq P$ a simplicial subset and $e \in \{0, 1\}$. Every map $P \times \{e\} \cup S \times \Delta[1] \to T$ extends to $P \times \Delta[1]$. In particular every open box of $\Delta[m] \times \Delta[1]^k$, $k \geq 1$, has a filler in T.*

Proof. Prisms over a simplex. A problem at n is a map $b : \Delta[n] \times \{e\} \cup Q \times \Delta[1] \to T$ with $Q \subseteq \Delta[n]$; its restriction along d_i is a problem at $n-1$. By recursion on n we choose a filler $F(b)$ with $F(b)\, d_i = F(b\, d_i)$. If Q contains the identity, $F(b) = b$. Otherwise the fillers $F(b\, d_i)$ agree on the common faces, since $F(b\, d_i)\, d_j = F(b\, d_i d_j)$, and $\partial\Delta[n] \times \Delta[1] \cup \Delta[n] \times \{e\}$ is given. The non-degenerate $(n+1)$-simplices of $\Delta[n] \times \Delta[1]$ are the shuffles

$$P_k = \big((0,0), \ldots, (k,0), (k,1), \ldots, (n,1)\big), \qquad 0 \leq k \leq n.$$

The face d_k of P_k is the face d_k of P_{k-1}, the face d_{k+1} of P_k is the face d_{k+1} of P_{k+1}, and the other faces of P_k lie in $\partial\Delta[n]\times\Delta[1]$, apart from $d_{n+1}P_n = \Delta[n]\times\{0\}$ and $d_0P_0 = \Delta[n]\times\{1\}$. For $e=0$ fill $P_n, P_{n-1}, \dots, P_0$ in this order: when P_k is filled, all its faces but d_k are known, so P_k is the filler of a horn on $\Lambda^k[n+1]$. For $e=1$ fill $P_0, \dots, P_n$, each on $\Lambda^{k+1}[n+1]$.

Prisms over a poset. For an injective $c : [n] \to P$ let b_c be the problem at n read along c, with $Q = c^*S$. Restriction along a face map gives $b_c\, d_i = b_{cd_i}$, so $F(b_c)\,\mu = F(b_{c\mu})$ for every injective μ, since these are composites of face maps. A simplex $g : [W] \to P$ factors uniquely as $g = c\sigma$ with c injective and σ surjective (Eilenberg–Zilber), and we put $F(g,j) = F(b_c)(\sigma, j)$. If $k : [W'] \to [W]$, the factorisation of gk is $(c\mu)\sigma'$ with μ injective and $\mu\sigma' = \sigma k$, so $F(gk, jk) = F(b_c)(\mu\sigma', jk) = F(g,j)\,k$. If $g \in S$ then $c \in S$, since $c = g\delta$ for a section δ of σ; the identity is then in c^*S, and $F(g,j) = b_c(\sigma, j)$ is the given value. The same holds at $j = e$.

Boxes. An open box of $\Delta[m]\times\Delta[1]^k$ is the union of all the codimension-one faces except one face $x_l = 1-e$ of the cube. Permuting the cube coordinates so that $l = k$, it is $P\times\{e\}\cup\partial P\times\Delta[1]$, where $P = [m]\times[1]^{k-1}$ and ∂P is the union of the faces of P. □

The fillers are uniform along the face maps only, so no excluded middle is used. The prism over P is filled simplex by simplex, without ordering the simplices of P. The shuffle order is used only inside each simplex. Let

$$L = \Sigma\,(u : \textstyle\prod_N \mathsf{D}_N A)\ \prod_N \mathrm{Path}_{\mathsf{D}_N A}(r\,u_{N+1}, u_N)$$

be the internal limit, and $\varphi : \mathsf{D}A \to L$ the map $u \mapsto ((r\,u)_N, \text{constant lines})$.

Theorem 4.2. *If A is fibrant, φ is a homotopy equivalence.*

Construction. Let $\ell = (u_N, p_N) \in L(X)$. For $f : [m] \to X$ put $\sigma_N = u_N(f)$, and let $p_N(f)$ be the prism $\Delta[m]\times\Delta[1] \to C_N$ obtained by reading p_N along $f\times[1]$ and extending by Eilenberg–Zilber, as in Theorem 4.1. We define, by recursion on m, an m-simplex $\psi(f)$ of C, and for every $N \geq m$:

- a prism $H^N(f) : \Delta[m]\times\Delta[1] \to C_N$ from $\psi(f)$ to σ_N;
- a square $\Theta_N(f) : \Delta[m]\times\Delta[1]^2 \to C_N$ with edges $H^{N+1}(f)$ and $H^N(f)$, $p_N(f)$, and the constant line at $\psi(f)$.

Each of these restricts along the face maps to the same datum at fd_i. For $N < m$ there is nothing to choose, since prisms and squares in C_N of an m-simplex with $m > N$ are determined by their faces. All the open boxes below are filled by Lemma 4.1. Each lies in dimensions $\leq m+1$, so in step 3, where $m+1 \leq N-1$, the box given in C_{N-1} is a box of C_N, and the image in C_{N-1} of the filled square is $\Theta_{N-1}(f)$. The boxes of steps 1 and 3 have dimension at most N, so they are boxes of C itself, and only step 2 uses C_m. Steps 1 and 3 alone define $\psi(\ell)$; step 2 is needed only for the homotopy.

1. $H^{m+1}(f)$ is the filler of the open box with lid σ_{m+1} and sides $H^{m+1}(fd_i)$. Its other end is $\psi(f)$. The box lies in dimensions $\leq m+1$, where C_{m+1} is C, so $\psi(f)$ is a simplex of C.
2. $H^m(f)$ and $\Theta_m(f)$ fill, in C_m, the open box of the square formed by $H^{m+1}(f)$, $p_m(f)$, the constant line, and the sides $\Theta_m(fd_i)$.
3. For $N \geq m+2$, $H^N(f)$ and $\Theta_{N-1}(f)$ fill, in C_N, the open box formed by $H^{N-1}(f)$, $p_{N-1}(f)$, the constant line, and the sides $\Theta_{N-1}(fd_i)$.

The data of fd_i used in each step have already been built. For fd_i, of dimension $m-1$, they come from its steps 1 and 3 with $N = m$ and $N = m+1$. The edges shared by a side and the rest of the box agree because they are themselves sides of earlier boxes. So $\psi(\ell) = (f \mapsto \psi(f))$ is natural

along the face maps, and is an element of $\mathsf{D}A(X)$. Each (H^N, Θ_N) is natural along the face maps, so it defines a line of $\mathsf{D}_N A$ and a square, read at the maps $(f, h) : [k] \to X \times [1]$ through $\langle \mathrm{id}, h \rangle$, as in Lemma 5.1. These form a path $\varphi\psi(\ell) \simeq \ell$ in L. Everything at f is computed from data along f, so the construction is natural in X and in Γ.

For $\psi\varphi \simeq \mathrm{id}$, let $u \in \mathsf{D}A(X)$ and run the construction on $\varphi(u)$. By the same recursion, using open boxes one dimension higher, build for each $f : [m] \to X$:

- a prism $G(f)$ in C from $\psi\varphi(u)(f)$ to $u(f)$, with sides $G(fd_i)$;
- for $N \geq m+1$, squares $\Xi^N(f)$ in C_N between $H^N(f)$ and $G(f)$, with constant top and bottom.

$G(f)$ and $\Xi^{m+1}(f)$ fill the box of $H^{m+1}(f)$ with sides $\Xi^{m+1}(fd_i)$. For $N \geq m+2$, $\Xi^N(f)$ fills, in C_N, the cube formed by $\Theta_{N-1}(f)$, $\Xi^{N-1}(f)$, constant faces, and the sides $\Xi^N(fd_i)$. Then G is a line of $\mathsf{D}A$ from $\psi\varphi(u)$ to u. □

Remark 4.3. Two features make the construction work.

- The value of $\psi(\ell)$ in dimension m is taken at stage $m+1$, never at a lower stage. In particular u_0 is never used, so the obstruction of Remark 3.3 does not arise.
- The squares Θ_N are part of the data and are built by box filling. They are not obtained by concatenating the p_N. A concatenation would not commute strictly with the faces: the face of $(H \cdot p^{-1}) \cdot p$ is not H.

The connectedness of the maps of the tower is witnessed by fillers of C and of the C_N. So no choice principle is used, whereas the internal argument from towers of connected maps needs dependent choice.

5 The propositional truncation

A type B is a proposition if any two elements w, w' of B over the same point have a path, natural in the stage: an element of B at $X \times [1]$ equal to w on $X \times \{0\}$ and to w' on $X \times \{1\}$. This is a section of $\Pi(b\, b' : B)\, \mathrm{Path}_B(b, b')$, read in the total presheaf.

Theorem 5.1. *Let A be fibrant. Then $\mathsf{D}_0 A$ is a fibrant D-modal proposition, and for every fibrant D-modal proposition B and map $g : A \to B$ there is a map $\mathsf{D}_0 A \to B$. So $A \to \mathsf{D}_0 A$ is the propositional truncation in the D-modal types.*

Proof. $\mathsf{D}_0 A$ is fibrant by Corollary 2.3, and modal since $\mathsf{D}C$ is modal whenever i^*C has a horn filling operation, which holds for $C = K_0 A$ (Lemma 1.2(3)). It is a proposition by Example 2.4: the path between u and v is u on $X \times \{0\}$ and v on $X \times \{1\}$.

A fibrant proposition B has sphere fillers, uniform along the face maps. For $w \in \mathsf{D}_0 B([k], \rho)$, that is $k+1$ vertices of B over a simplex ρ of the context, define $\mathrm{fl}(w) \in B([k], \rho)$ with faces $\mathrm{fl}(w\, d_i)$, by recursion on k. For $k = 0$ it is the vertex. For $k+1$ the faces $s_i = \mathrm{fl}(w\, d_i)$ form a sphere. A filler c of the horn $(s_i)_{i \geq 1}$ (Lemma 2.1) lies over ρ; the box over $[k+1]$ with lid c at 1 and, on the boundary, the paths of B from the sphere to c is filled by Lemma 4.1, and $\mathrm{fl}(w)$ is its other end. Then $f \mapsto \mathrm{fl}(wf)$ is a map $\mathsf{D}_0 B \to \mathsf{D}B$, natural since fl depends only on wf. The map $\mathsf{D}_0 A \to B$ is $p_B \circ \mathrm{fl} \circ \mathsf{D}_0 g$, with p_B an inverse of η_B. The homotopies needed for the universal property are free, since B is a proposition. □

For $n \geq 0$ the same pattern needs sphere fillers in an n-truncated type for spheres of dimension $\geq n+1$, which requires relating iterated path types to spheres (Section 6, item 1). The alternative

induction through $\mathrm{Path}_{\mathsf{D}_{n+2}A}(\eta a, \eta b) \simeq \mathsf{D}_{n+1}\mathrm{Path}_{\mathsf{D}A}(\eta a, \eta b)$ needs, besides this comparison (a homotopy equivalence, not an isomorphism: the $(n+2)$-simplices of $\mathsf{D}_{n+2}A$ over a non-degenerate $(n+2)$-simplex of X have no counterpart), the internal lemmas on connected maps, for which the model has no internal language.

6 What remains

The statement *every* D*-modal type is the limit of its Postnikov tower* needs the following points beyond Sections 1–4.

1. The comparison between the internal truncation level (iterated identity types are propositions) and the unique filling of spheres of Proposition 3.1, for fibrant types of the poset model.
2. The universal property of $A \to \mathsf{D}_{n+1}A$ among the D-modal n-truncated types B. A natural route is to show that $\mathsf{D}B \to \mathsf{D}_{n+1}B$ is a homotopy equivalence for such B, since spheres of dimension $> n$ bound in B, and then to argue as in Proposition 6.1. That argument also needs $\eta_{\mathsf{D}_N A} \simeq \mathsf{D}_N \eta_A$, and the homotopy of Lemma 5.2 does not restrict to D_N: it reads u at dimensions up to $|B_0| + |A_1| - 1$, which may exceed N.
3. Whitehead's principle in the form: if $\alpha : A \to B$ is a map of fibrant types and every $\mathsf{D}_N\alpha$ is a homotopy equivalence, then so is $\mathsf{D}\alpha$. With Theorem 4.2 it reduces to the invariance of the internal limit under a levelwise equivalence of towers. That invariance is a statement of type theory with Π, Σ and Path, and needs no property of the model.

A Correspondence with the formalisation

Layer L9 of `seg/core`, over the poset instance.

statement	files
$K_N A$, η, the restrictions	CCosk (KT, Res)
Lemma 1.2(3)	CCoskHorn (HK; the count: pigN, InHorn); CLocHorn now reads a horn through its faces (FD, HSF, HPF, FaceFill), and HornOfFib is the instance for i^*A
Definition of D_N, Corollary 2.3	CLocDN (DN.fib = Theorem 4.1 for $K_N A$); naturality: CCoskCart, CLocDNUniv
Proposition 2.2	CLocDNC (to, from, to-from, from-to, to-nat); CLocEZ.sN-le, r-le
Proposition 3.2	CLocDN (Lim: u, RN.res-u, uniq)
Lemma 4.1	CBoxFill (BF: the problem at c, RS, the filled prisms fl, the map Fv with Fp, Fn, Fe, Fs); the factorisation: CBoxEZ (EZX, EZXK, EZXI; heights HLin, HProd); faces suffice: CFacesGen (CLocD.FacesGen is its instance); squares: CBoxSq
Theorem 4.2, the construction of ψ	reading u_N, p_N along f: CPostMap (DR, DRI, DRZ, DRP, DRId); sides of the boxes: CPostBdry (BG, BS; CFaceTwo, shared with CLocHorn); the open box of a square over a simplex, for any type with face-uniform fillers: CPostSq (SqE, SqS, SqZ); step 1: CPostStep1 (P1, S1); step 3: CPostStep3 (E3; S3, instance of SqS); dimension 0: CPostStep0; the recursion: CPostRec (Rec, SRec over the stages, ST from m to $m+1$, FFS the agreement of faces, Drec, psi $\in \mathsf{D}A(X)$)
Theorem 5.1	IsPropE, D0P (D_0A a proposition), D0M (fibrant, modal; CLocUniv.dModalH, CLocEq.FibD): CTruncProp; SF (sphere filling fl, the map sN : $\mathsf{D}_0 B \to \mathsf{D}B$), KMap ($K_N$ on maps), UP0.k ($\mathsf{D}_0 A \to B$): CTruncFill
Theorem 4.2, $\varphi\psi \simeq \mathrm{id}$	η and the value at the identity (Lemma 1.2(1)): CCosk (etaE, ev, eta-ev, ev-eta); reading p_N along $f \times [1]$ in $K_N A$: CPostK (DRK, DRKI, DRKE; along strict chains DRS, DRKS); step 2: CPostStep2 (E2; S2, S2z, instances of SqS, SqZ for $B = K_m A$ with the fillers of CCoskHorn); faces suffice below a bound: CFacesGen.SGB (SG is its instance); the lines: CPostLine (St2S, St2Z: step 2 at every simplex; the family val, natural along faces WF, along strict maps FG; DCN.line $\in \mathsf{D}_N A(X \times [1])$; end0, end1); the squares: CPostSquare (val2, DCS.sq $\in \mathsf{D}_N A((X \times [1]) \times [1])$; edges sq-s0, sq-s1, sq-t0, sq-t1)

Not formalised: Lemma 1.2(2) and Proposition 3.1, Example 2.4 (except $\mathsf{D}_0 A$ a proposition), Section 6; of Theorem 4.2, the homotopy $\psi\varphi \simeq \mathrm{id}$ and the naturality of ψ and of the homotopy in X. For $\varphi\psi \simeq \mathrm{id}$ the lines and squares are formalised with all their edges, at a fixed (X, ℓ); the internal limit L is not defined as a type, and the path in L is given by these components.